$\Delta = b^2 - 4ac$

$x = \frac{-b \pm \sqrt{\Delta}}{2a}$

Trabalho de cálculo

1 A) $f(x) = \log(2x-6) \rightarrow 2x - 6 > 0$

$2x > 6$

$x > 3 \rightarrow Dom(f) \{x \in \mathbb{R} \mid x > 3\}$

B) $f(x) = \frac{\sqrt{6-x^2-x}}{x^2+3x+4} \rightarrow x^2 + 3x + 4 \neq 0$

$\Delta = 3^2 - 4.4 = -7$

$6 - x^2 - x \geq 0$

$\Delta = -1^2 - 4.(-1).6 = 25$

$x_{+} = \frac{-(-1) \pm \sqrt{25}}{2.(-1)}$

$x_{I} = \frac{1+5}{-2} = -3$

$x_{II} = \frac{1-5}{-2} = 2$

$[-3, 2]$

C) $f(x) = \frac{3x^2 - 5x + 7}{\ln(x^2-36)} \rightarrow x^2 - 36 > 0$

$\Delta = -4.(-36) = 144$

$x_{I} = \frac{+\sqrt{144}}{2} = \frac{12}{2} = 6$

$x_{II} = \frac{-\sqrt{144}}{2} = \frac{-12}{6} = -6$

$\ln(x^2-36) \neq 0$

$x^2 - 36 = 1$

$x^2 = +37$

$\sqrt{x^2} = \sqrt{37}$

$|x| \neq 37$

$Dom(f) = \{x \in \mathbb{R} \mid x < -6 \text{ ou } x > 6, x \neq \sqrt{37}, x \neq -\sqrt{37}\}$

D) $\frac{\cos x}{\operatorname{sen}(3x)} \rightarrow \operatorname{sen}(3x) = 0$ quando $3x\ m\pi$ logo

$Dom(f) = \{x \in \mathbb{R} \mid x \neq 0 \text{ e } x \neq \frac{m\pi}{3} \text{ para todo } m \in \mathbb{Z}\}$

$Dom(f) = \{x \in \mathbb{R} \mid x \neq \frac{m\pi}{3}, m \in \mathbb{Z}\}$ → considere isto

E) $\sqrt{x+1} \longrightarrow x \geq -1$

$\cos\left(x+\frac{\pi}{2}\right) \longrightarrow \cos\left(x+\frac{\pi}{2}\right) \neq 0$ $\quad$ $\cos(\theta) = 0$ quando $\theta = \frac{\pi}{2} + m\pi$

$x + \frac{\pi}{2} = \frac{\pi}{2} + m\pi$

$x = m\pi$ $\quad$ $(m \in \mathbb{Z})$

$Dom(f) = \{x \in \mathbb{R} \mid x \neq m\pi \text{ e } x \geq -1, m \in \mathbb{Z}\}$

f) $f(x) = \dfrac{4+e^x}{\sqrt{x^2-7x-8}} \rightarrow x^2 - 7x - 8 > 0$

$\Delta = (-7)^2 - 4(-8) = 81$

$a = 1 > 0$ $\quad$ $x = \dfrac{7 \pm 9}{2}$

$Dom(f) = (-\infty, -1) \cup (8, +\infty)$ $\quad$ $x_I = 8$

$x_{II} = -1$

G) $f(x) = \dfrac{3x+4}{2^x - 1} \rightarrow 2^x - 1 \neq 0 \rightarrow 2^x \neq 1 \rightarrow 2^x \neq 2^0 \rightarrow x \neq 0$

$Dom(f) = \{x \in \mathbb{R} \mid x \neq 0\}$

Questão 2:

(a) $f(x) = 5x - 3$

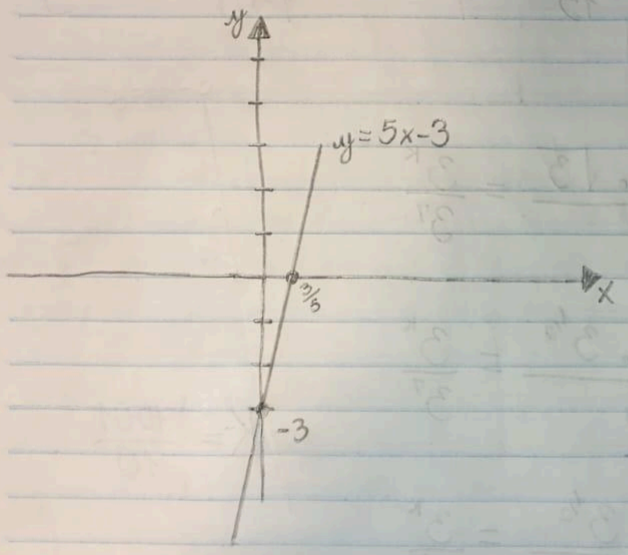
$x = 0 \rightarrow y = -3$

$y = 0 \rightarrow 5x = 3$

$x = \frac{3}{5}$

$Dom(f) = \mathbb{R}$

$Im(f) = \mathbb{R}$



(b) $f(x) = x^2 + 4x - 21$

$\Delta = (4)^2 - 4 . (1) . (-21)$

$\Delta = 16 + 84$

$\Delta = 100$

$Dom(f) = \mathbb{R}$

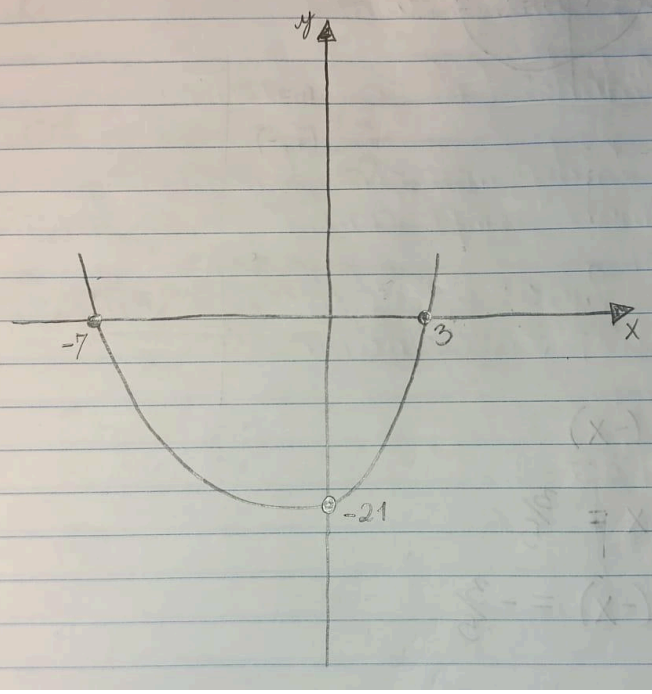
$Im(f) = [-25, +\infty[$

$$\frac{-4 \pm \sqrt{100}}{2 \cdot 1}$$

$$\frac{-4 \pm 10}{2} = \begin{cases} x' = 3 \\ x'' = -7 \end{cases}$$

$$yV = \frac{-100}{4} = -25$$

$$x = -21$$

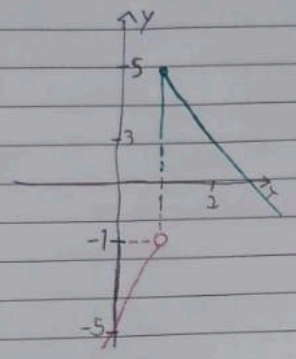


2.c)

$$f(x) = \begin{cases} -2x^2+6x-5 & \text{se } x < 1 \\ 7-2x & \text{se } x \geq 1 \end{cases}$$

$x_v = \frac{-6}{2.(-2)} = 1.5$

$f(1^-) = -2(1)^2+6(1)-5 = -1$



$\Delta = 6^2 - 4(-2).-5 = -4$ $(-\infty, -1)$

$7-2(1) = 5$ $(-\infty, 5]$

$Dom(f) = \{x \in \mathbb{R}\}$

$Im(f) = (-\infty, 5]$

D) $f(x) = \begin{cases} 3^x & \text{se } x \leq 0 \\ 2x+1 & \text{se } 0 < x < 3 \\ x^2-8x+16 & \text{se } x \geq 3 \end{cases}$

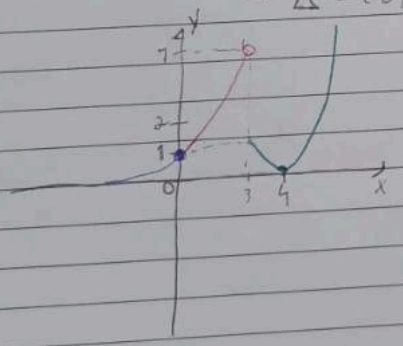
$\longrightarrow Im(0, 1]$

$\longrightarrow 2.(0)+1 = 1$

$2(3)+1 = 7$

$Im(f)(1, 7)$

$\Delta = (-8)^2 - 4.16 = 0$



$x_v \ \frac{-(-8)}{2.1} = 4, y_v = 0$

$3^2 - 8.(3) + 16 = 1$

$Dom(f) = \{x \in \mathbb{R}\}$

$Im(f) = [0; \infty)$

Questão 3:

$$\frac{3^{81} \cdot 27^{4} \cdot \sqrt{3}}{\sqrt[5]{9}} = \frac{3^{K}}{3^{7}}$$

$$\frac{3^{81} \cdot (3^{3})^{4} \cdot 3^{1/2}}{(3^{2})^{1/5}} = \frac{3^{K}}{3^{7}}$$

$$\frac{3^{81} \cdot 3^{12} \cdot 3^{1/2}}{3^{2/5}} = \frac{3^{K}}{3^{7}}$$

$$\frac{3^{\frac{187}{2}}}{3^{2/5}} = \frac{3^{K}}{3^{7}}$$

$$3^{\frac{931}{10}} \cdot 3^{7} = 3^{K}$$

$$K = \frac{1001}{10}$$

Questão 4:

$$3 - \log_5 10 + \log_5 7 - \log_5 15 - 3\log_{1/5} 4 = \log_5 K$$

$$\log_5 (5^3) - \log_5 10 + \log_5 7 - \log_5 15 + \log_5 (4^3) = \log_5 K$$

$$\log_5 \left(\frac{125}{10}\right) + \log_5 \left(\frac{7}{15}\right) + \log_5 64 = \log_5 K$$

$$K = \frac{125}{10} \cdot \frac{7}{15} \cdot \frac{64}{1} = \frac{56000}{150} = \frac{1120}{3}$$

Questão 5:

(a) $\cos x$

$\operatorname{sen}^2 x + \cos^2 x = 1$

$\left(\frac{3}{5}\right)^2 + \cos^2 x = 1$

$\cos^2 x = 1 - \frac{9}{25}$

$\cos^2 x = \frac{16}{25}$

$\cos x = -\frac{4}{5}$

(b) $\operatorname{sen}(-x)$

$\operatorname{sen} x = 3/5$

$\operatorname{sen}(-x) = -3/5$

(c) $\cos(-2x)$

$\cos(2x) = \cos^2 x - \operatorname{sen}^2 x$

$\cos(2x) = \left(-\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2$

$\cos(2x) = \frac{16}{25} - \frac{9}{25}$

$\cos(2x) = \frac{7}{25}$

(d) $\cos\left(x - \frac{\pi}{4}\right)$

$$\cos\left(x - \frac{\pi}{4}\right) = \cos x \cdot \cos\frac{\pi}{4} + \operatorname{sen} x \cdot \operatorname{sen}\frac{\pi}{4}$$

$$\cos\left(x - \frac{\pi}{4}\right) = \left(-\frac{4}{5} \cdot \frac{\sqrt{2}}{2}\right) + \left(\frac{3}{5} \cdot \frac{\sqrt{2}}{2}\right)$$

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{-4\sqrt{2}}{10} + \frac{3\sqrt{2}}{10}$$

$$\cos\left(x - \frac{\pi}{4}\right) = -\frac{\sqrt{2}}{10}$$